Avaliacao de *solvers* de programacao nao-linear de codigo aberto sobre os *benchmarks* AMPL-NLP e COCONUT por meio do JuMP.jl

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Abstract

A escolha de um *solver* adequado e uma etapa decisiva na solucao de problemas de programacao nao-linear (NLP), pois diferentes algoritmos apresentam desempenho e robustez distintos conforme a estrutura do problema. Este trabalho reproduz e estende dois conjuntos publicos de teste amplamente utilizados — o *benchmark* AMPL-NLP de H. Mittelmann e a Biblioteca 1 do *benchmark* COCONUT — empregando *solvers* de codigo aberto acessiveis pelo ecossistema JuMP.jl: Ipopt, MadNLP, Uno, NLopt e Optim. Os modelos foram obtidos em formato `.mod` (AMPL) e convertidos para `.nl`; cinco dos 47 arquivos da Parte 1 nao puderam ser convertidos por nao serem modelos de otimizacao, restando 41 problemas utilizaveis. Para a Parte 2 foram selecionados, da Biblioteca 1 do COCONUT, os 202 problemas com solucao de referencia conhecida, dos quais 201 foram efetivamente convertidos. A execucao sistematica concentrou-se em Ipopt e MadNLP, com tempo limite de 500 s (Parte 1) e 300 s (Parte 2). NLopt e Optim mostraram-se inadequados a esta classe de problemas, e o Uno foi viavel apenas em um caso isolado. Na Parte 1, Ipopt resolveu 25 e MadNLP resolveu 26 dos 41 problemas, com Ipopt em media 2,4 vezes mais rapido. Na Parte 2, Ipopt resolveu 192 e MadNLP 188 dos 201 problemas, com *gap* mediano da ordem de $10^{-4}\%$ em relacao a melhor solucao conhecida.

Keywords: programacao nao-linear, *solvers* de otimizacao, *benchmark*, JuMP, AMPL, *gap* de otimalidade

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1. Conclusao

Este relatorio apresenta os resultados da execucao sistematica dos *solvers* de codigo aberto Ipopt e MadNLP sobre os *benchmarks* AMPL-NLP (Parte 1) e COCONUT — Biblioteca 1 (Parte 2). Os experimentos foram conduzidos em um notebook Dell Inspiron 15 5510, com processador Intel Core i7-11390H @ 3.40 GHz (4 nucleos fisicos, 8 *threads*) e 16 GB de memoria RAM, sob Windows 11 Pro 64 bits, utilizando Julia 1.12.5 com o JuMP.jl como camada de modelagem e os pacotes Ipopt.jl, MadNLP.jl, Uno.jl, NLopt.jl e Optim.jl como *solvers*.

Em testes preliminares observou-se que os *solvers* NLopt e Optim nao sao adequados a esta classe de problemas: o NLopt apresenta dificuldade com restricoes nao-lineares gerais (sendo projetado sobretudo para restricoes de caixa) e o Optim nao aceita modelos no formato `.nl`. O *solver* Uno chegou a uma solucao valida apenas em um problema testado (`ex8_2_3`), em tempo cerca de 65 vezes superior ao do Ipopt, e em testes subsequentes ultrapassou o limite de tempo sem retornar solucao, alem de nao respeitar o parametro `set_time_limit_sec` do JuMP. Por essas razoes, a analise sistematica restringiu-se a Ipopt e MadNLP.

O *gap* de otimalidade utilizado na Parte 1 foi

$$\text{gap} = \frac{|f_{\text{obtido}} - f_{\text{ref}}|}{1 + |f_{\text{ref}}|}, \tag{1}$$

e na Parte 2, em forma percentual,

$$\text{gap}(\%) = \frac{|f_{\text{obtido}} - F_{\text{best}}|}{|F_{\text{best}}|} \times 100\%, \tag{2}$$

adotando-se o *gap* absoluto nos casos em que $F_{\text{best}} \approx 0$.

Parte 1 – AMPL-NLP.. Dos 47 problemas do *benchmark*, 5 nao puderam ser convertidos por nao serem modelos de otimizacao (arquivos HTML), restando 41 problemas utilizaveis. O Ipopt resolveu 25 deles, o MadNLP resolveu 26, e ambos atingiram o limite de tempo (500 s) em 14 e 13 casos, respectivamente, alem de 2 ocorrencias de limite de iteracoes em cada *solver*. Os valores da funcao-objetivo obtidos pelos dois *solvers* concordaram em pelo menos quatro casas decimais na grande maioria dos casos, com unica excecao relevante o problema `NARX_CFy` (diferenca de aproximadamente 1,46%). Considerando os 25 problemas resolvidos por ambos, o Ipopt foi mais rapido em 20 casos, com tempo medio de 111 s contra 272 s do MadNLP — ou seja, em media 2,4 vezes mais rapido. O caso mais extremo foi o `gasoil_3200`, em que Ipopt convergiu em 17 s e MadNLP em 3174 s. Os problemas `robot_b` e `robot_c` atingiram o limite de iteracoes em ambos os *solvers*, indicando dificuldade estrutural desses modelos. A Tabela 1 apresenta os resultados completos da Parte 1.

Parte 2 – COCONUT, Biblioteca 1.. A partir da tabela da Biblioteca 1 foram selecionados os problemas com `modestatus` = 2, ou seja, com solucao localmente otima conhecida (F_{best} definido), totalizando 202 problemas. Destes, 201 foram convertidos com sucesso para o formato `.nl` — o problema `robot50` falhou na conversao por inconsistencia no arquivo `.mod` original. Aplicou-se tempo limite de 300 s por execucao. O Ipopt resolveu 192 problemas (96%) e o MadNLP resolveu 188 (94%), com *gap* mediano de 0,00025% e 0,00042%, respectivamente. O numero de problemas com *gap* inferior a 0,01% foi 130 (Ipopt) e 120 (MadNLP). Cerca de 14% dos problemas resolvidos apresentaram *gap* superior a 10%, comportamento esperado uma vez que tanto Ipopt quanto MadNLP sao *solvers* de otimizacao local e podem convergir, em problemas com multiplos otimos, para uma solucao valida diferente do otimo global indicado no *benchmark*. A Tabela 2 apresenta os resultados completos da Parte 2.

Legenda de status *(Tabelas 1 e 2):.* OK = `LOCALLY_SOLVED`; OK* = `ALMOST_LOCALLY_SOLVED`; TIME = `TIME_LIMIT`; ITER = `ITERATION_LIMIT`; NO NL = arquivo `.nl` nao gerado; INV = `INVALID_MODEL`; NUM = `NUMERICAL_ERROR`; INF = `LOCALLY_INFEASIBLE`; ERR = `OTHER_ERROR`; SLOW = `SLOW_PROGRESS`. "–" indica valor nao disponivel (*e.g.* sem objetivo por interrupcao, ou sem registro de iteracoes na rodada).

Limitacoes e trabalhos futuros.. Destacam-se a sensibilidade ao limite de tempo nos problemas de maior porte da Parte 1 e a necessidade de licenca/instalacao do AMPL para a conversao dos modelos. Como trabalhos futuros, sugere-se reexecutar a Parte 1 com Uno apos as proximas versoes do *solver*, ajustar tolerancias e limites de tempo, e investigar o uso de estrategias de *multi-start* para os problemas da Parte 2 com *gap* elevado.

Table 1: Resultados completos do *benchmark* AMPL-NLP (Parte 1). Tempo em segundos; "it." = iteracoes; "var." = variaveis; "rest." = restricoes.

			Ipopt				**MadNLP**			
Problema	var.	rest.	status	tempo	it.	obj.	status	tempo	it.	obj.
NARX_CFy	43973	0	OK	175.78	388	0.0086	OK	229.20	481	0.0087
WM_CFy	8709	1512	TIME	500.00	–	–	TIME	500.00	–	–
arki0003	1872	2108	TIME	500.00	–	–	TIME	500.00	–	–
arki0009	6220	6131	TIME	500.00	–	–	TIME	500.00	–	–
bearing_400	160000	160000	TIME	500.00	–	–	TIME	500.00	–	–
camshape_6400	6400	12800	OK	8.51	–	4.4484	OK	28.54	–	4.4484
clnlbeam	59999	79996	OK	153.51	–	344.8761	OK	501.52	–	344.8761
cont5_1_l	90600	181200	OK	14.36	16	2.7210	OK	24.42	16	2.7210
cont5_2_1_l	90600	181200	OK	36.54	39	6.628e-4	OK	54.69	41	6.627e-4
cont5_2_2_l	90600	181200	OK	42.80	45	5.200e-4	OK	65.46	48	5.200e-4
cont5_2_3_l	90600	181200	OK	53.12	56	5.208e-4	OK	74.83	55	5.208e-4
cont5_2_4_l	90600	600	OK	35.43	14	0.0664	OK	94.33	11	0.0664
corkscrw	44997	40000	OK	57.72	–	98.0960	OK	34.64	–	98.0960
dirichlet_120	–	–	NO NL	–	–	–	NO NL	–	–	–
dtoc1nd	5960	0	OK	32.28	8	136.4558	OK	25.69	8	136.4558
dtoc2	64950	0	TIME	500.00	–	–	TIME	500.00	–	–
elec_400	1200	0	OK	1425.38	–	7.558e+4	OK	1700.60	–	7.558e+4
ex1_160	50562	101124	OK	6.55	–	0.0639	OK	8.30	–	0.0639
ex1_320	203522	407044	OK	22.76	9	0.0654	OK	38.32	8	0.0654
ex4_2_160	51198	102396	OK	13.21	–	3.6461	OK	18.09	–	3.6461
ex4_2_320	204798	409596	OK	50.37	17	3.6392	OK	68.22	17	3.6392
ex8_2_2	7510	15020	TIME	500.00	–	–	TIME	500.00	–	–
ex8_2_3	15636	31272	OK	3.54	–	-3731.0793	OK	3.70	–	-3731.0793
gasoil_3200	32001	3	OK	17.72	–	0.0052	OK	3174.11	–	0.0052
henon_120	–	–	NO NL	–	–	–	NO NL	–	–	–
lane_emden_120	–	–	NO NL	–	–	–	NO NL	–	–	–
nql180	–	–	NO NL	–	–	–	NO NL	–	–	–
optmass	60006	0	OK	7.83	–	-0.1204	OK	14.18	–	-0.1204
pinene_3200	64000	5	OK	8.50	13	19.8722	OK	8.02	12	19.8722
qcqp1000-1nc	1000	0	OK	251.92	–	-2.663e+7	OK	248.51	–	-2.663e+7
qcqp1000-2c	1000	0	OK	272.14	101	7.381e+5	OK	281.56	95	7.381e+5
qcqp1000-2nc	1000	0	TIME	501.87	344	9.348e+4	OK	211.32	142	9.328e+4
qcqp1500-1c	1500	0	TIME	500.00	–	–	TIME	500.00	–	–

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(continuacao da Tabela 1)

			Ipopt				MadNLP			
Problema	var.	rest.	status	tempo	it.	obj.	status	tempo	it.	obj.
qcqp1500-1nc	1500	0	TIME	500.00	–	–	TIME	500.00	–	–
qcqp500-3c	500	0	TIME	500.00	–	–	TIME	500.00	–	–
qcqp500-3nc	500	0	TIME	500.00	–	–	TIME	500.00	–	–
qcqp750-2c	750	0	TIME	500.00	–	–	TIME	500.00	–	–
qcqp750-2nc	750	0	TIME	500.00	–	–	TIME	500.00	–	–
qssp180	–	–	NO NL	–	–	–	NO NL	–	–	–
robot_1600	14399	19201	OK	22.83	–	9.1409	OK	36.18	–	9.1409
robot_a	1001	0	TIME	500.00	–	–	TIME	500.00	–	–
robot_b	1001	0	ITER	6226.70	–	15.5185	ITER	31042.99	–	19.9890
robot_c	1001	0	ITER	7536.00	–	30.8256	ITER	6356.37	–	16.2904
rocket_12800	51201	76801	OK	15.75	–	1.0128	OK	16.59	–	1.0128
steering_12800	64000	25603	OK	15.79	–	0.5546	OK	20.46	–	0.5546
svanberg	50000	100000	OK	32.98	34	8.362e+4	OK	24.44	34	8.362e+4

Table 2: Resultados completos do *benchmark* COCONUT — Biblioteca 1 (Parte 2). Tempo em segundos; *gap* em percentual (%). F_{best} = melhor valor conhecido (referencia).

		Ipopt				MadNLP			
Problema	F_{best}	status	tempo	obj.	gap (%)	status	tempo	obj.	gap (%)
abel	225.1946	OK	6.77	225.1946	7.00e-06	OK	12.47	225.1946	7.00e-06
alkyl	-1.7650	OK	0.05	-1.7650	1.00e-05	OK	0.15	-1.7650	1.00e-05
camshape100	-4.2841	OK	0.05	-4.2842	0.0023	OK	0.02	-4.2842	0.0023
camshape200	-4.2785	OK	0.11	-4.2787	0.0048	OK	2.94	-4.2787	0.0048
catmix100	-0.0481	OK	0.03	-0.0481	0.0637	OK	0.02	-0.0481	0.0638
catmix200	-0.0481	OK	0.03	-0.0481	0.0854	OK	0.04	-0.0481	0.0854
chain100	5.0698	OK	0.02	5.0698	3.04e-04	OK	0.01	5.0698	3.04e-04
chain200	5.0689	OK	0.01	5.0689	3.42e-04	OK	0.01	5.0689	3.42e-04
chain400	5.0686	OK	0.02	5.0686	4.28e-04	OK	0.02	5.0686	4.28e-04
chain50	5.0723	OK	0.01	5.0723	7.59e-04	OK	0.00	5.0723	7.59e-04
chakra	-179.1336	OK	0.01	-179.1336	2.40e-05	OK	0.00	-179.1336	2.40e-05
chance	29.8944	OK	0.00	29.8944	7.30e-05	OK	0.00	29.8944	7.30e-05
chem	-47.7065	OK	0.01	-47.7065	3.10e-05	OK	0.01	-47.7065	3.10e-05
chenery	-1058.9199	OK	0.01	-1058.9199	4.00e-06	OK	0.00	-1058.9199	4.00e-06
circle	4.5742	OK	0.01	4.5742	0.0010	OK	0.00	4.5742	0.0010
demo7	-1.589e+6	OK	0.06	-1.589e+6	0.00e+00	OK	0.02	-1.589e+6	0.00e+00
dispatch	3155.2879	OK	0.01	3155.2879	0.00e+00	OK	0.00	3155.2879	0.00e+00
elec25	243.8127	OK	0.10	243.8128	2.50e-05	OK	0.08	243.8128	2.50e-05
elec50	1055.1820	OK	1.29	1055.1823	3.00e-05	OK	1.03	1055.1823	3.00e-05
etamac	-15.2947	OK	0.03	-15.2947	1.59e-04	OK	0.01	-15.2947	1.59e-04
ex14_1_1	0.0000	OK	0.00	-1.346e-7	0.00e+00	OK	0.00	-1.346e-7	0.00e+00
ex14_1_2	0.0000	OK	0.01	1.004e-8	0.00e+00	OK	0.00	1.004e-8	0.00e+00
ex14_1_3	0.0000	OK	0.01	1.862e-11	0.00e+00	OK	0.00	1.862e-11	0.00e+00
ex14_1_4	0.0000	OK	0.01	-7.505e-9	0.00e+00	OK	0.00	-8.140e-9	0.00e+00
ex14_1_5	0.0000	OK	0.02	-4.982e-9	0.00e+00	OK	0.00	-4.988e-9	0.00e+00
ex14_1_6	0.0000	OK*	0.21	1.0000	1.00	OK*	0.12	1.0000	1.00
ex14_1_7	0.0000	OK	0.03	0.1350	0.1350	OK	0.02	0.1350	0.1350
ex14_1_8	0.0000	OK	0.01	0.0414	0.0414	OK	0.00	0.0414	0.0414
ex14_1_9	0.0000	OK	0.01	-4.984e-9	0.00e+00	OK	0.00	-4.985e-9	0.00e+00
ex14_2_1	0.0000	OK	0.01	-1.549e-9	0.00e+00	OK	0.01	5.657e-10	0.00e+00
ex14_2_2	0.0000	OK	0.00	-4.511e-9	0.00e+00	OK	0.00	-5.482e-9	0.00e+00
ex14_2_3	0.0000	OK	0.01	-6.773e-9	0.00e+00	OK	0.00	-8.916e-9	0.00e+00
ex14_2_4	0.0000	OK	0.00	-1.410e-10	0.00e+00	OK	0.00	-2.216e-10	0.00e+00
ex14_2_5	0.0000	OK	0.01	-7.192e-9	0.00e+00	OK	0.00	-7.140e-9	0.00e+00
ex14_2_6	0.0000	OK	0.01	-9.026e-9	0.00e+00	OK	0.00	-9.016e-9	0.00e+00
ex14_2_7	0.0000	OK	0.01	-9.133e-9	0.00e+00	OK	0.00	-8.792e-9	0.00e+00
ex14_2_8	0.0000	OK	0.02	-7.227e-9	0.00e+00	OK	0.00	-9.027e-9	0.00e+00
ex14_2_9	0.0000	OK	0.01	-5.443e-9	0.00e+00	OK	0.00	-5.489e-9	0.00e+00
ex2_1_1	-17.0000	OK	0.01	-17.0000	1.50e-05	OK	0.00	-17.0000	1.50e-05
ex2_1_10	4.932e+4	OK	0.04	4.932e+4	1.60e-05	OK	0.01	4.932e+4	1.60e-05
ex2_1_2	-213.0000	OK	0.01	-213.0000	2.00e-06	OK	0.00	-213.0000	2.00e-06
ex2_1_3	-15.0000	OK	0.01	-15.0000	2.00e-06	OK	0.00	-15.0000	2.00e-06
ex2_1_4	-11.0000	OK	0.01	-11.0000	2.00e-06	OK	0.00	-11.0000	2.00e-06
ex2_1_5	-268.0146	OK	0.01	-268.0146	1.40e-05	OK	0.00	-268.0146	1.40e-05
ex2_1_6	-39.0000	OK	0.01	-39.0000	1.30e-05	OK	0.00	-39.0000	1.30e-05
ex2_1_7	-4150.4101	OK	0.02	-4150.4103	4.00e-06	OK	0.00	-4150.4103	4.00e-06
ex2_1_8	1.564e+4	OK	0.03	1.564e+4	1.00e-06	OK	0.01	1.720e+4	9.95
ex2_1_9	-0.3750	OK	0.01	-0.3750	3.00e-06	OK	0.00	-0.3750	3.00e-06

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(continuacao da Tabela 2)

		Ipopt				MadNLP			
Problema	F_{best}	status	tempo	obj.	gap (%)	status	tempo	obj.	gap (%)
ex3_1_1	7049.2083	OK	0.00	7049.2479	5.62e-04	OK	0.00	7049.2478	5.60e-04
ex3_1_2	-3.067e+4	OK	0.01	-3.067e+4	4.00e-06	OK	0.01	-3.067e+4	4.00e-06
ex3_1_3	-310.0000	OK	0.02	-310.0000	3.00e-06	OK	0.01	-310.0000	3.00e-06
ex3_1_4	-4.0000	OK	0.01	-4.0000	4.00e-06	OK	0.00	-4.0000	4.00e-06
ex4_1_1	-7.4873	OK	0.01	-0.5200	93.06	OK	1.03	-0.5200	93.06
ex4_1_2	-663.5001	OK	0.00	-663.5001	1.00e-06	OK	0.00	-663.5001	1.00e-06
ex4_1_3	-443.6717	OK	0.00	-443.6717	1.00e-06	OK	0.00	-443.6717	1.00e-06
ex4_1_4	0.0000	OK	0.00	0.0000	0.00e+00	OK	0.00	0.0000	0.00e+00
ex4_1_5	0.0000	OK	0.00	7.176e-20	0.00e+00	OK	0.03	7.176e-20	0.00e+00
ex4_1_6	7.0000	OK	0.01	7.0000	0.00e+00	SLOW	0.00	7.0000	0.00e+00
ex4_1_7	-7.5000	OK	0.00	-7.5000	0.00e+00	OK	0.00	-7.5000	0.00e+00
ex4_1_8	-16.7389	OK	0.01	-16.7389	4.10e-05	OK	0.00	-16.7389	4.10e-05
ex4_1_9	-5.5080	OK	0.00	-5.5080	2.46e-04	OK	0.00	-5.5080	2.46e-04
ex5_2_2_case1	-400.0000	OK	0.02	-400.0000	1.00e-06	OK	0.01	-100.0000	75.00
ex5_2_2_case2	-600.0000	OK	0.03	-400.0000	33.33	OK	0.01	-400.0000	33.33
ex5_2_2_case3	-750.0000	OK	0.02	-750.0000	1.00e-06	OK	0.01	-750.0000	1.00e-06
ex5_2_4	-450.0000	OK	0.01	-450.0000	1.00e-06	OK	0.01	-450.0000	1.00e-06
ex5_2_5	-3500.0001	OK	0.17	-3500.0000	1.00e-06	OK	1.84	-3500.0000	1.00e-06
ex5_3_2	1.8642	OK	0.05	1.8642	0.0022	OK	0.02	1.8642	0.0022
ex5_3_3	3.2340	OK	0.42	3.8251	18.28	INF	0.20	3.9240	21.34
ex5_4_2	7512.2259	OK	0.01	7512.2300	5.50e-05	OK	0.00	7512.2299	5.30e-05
ex5_4_3	4845.4620	OK	0.02	4845.4620	0.00e+00	OK	0.02	4845.4620	0.00e+00
ex5_4_4	1.008e+4	OK	0.04	1.214e+4	20.45	OK	0.03	1.184e+4	17.50
ex6_1_1	-0.0202	OK	0.01	-0.0176	12.98	OK	0.01	-0.0176	12.98
ex6_1_2	-0.0325	OK	0.01	-0.0325	0.1115	OK	0.00	-0.0325	0.1115
ex6_1_3	-0.3525	OK	0.02	-0.3243	7.99	OK	0.01	-0.3243	7.99
ex6_1_4	-0.2945	OK	0.01	-0.2945	0.0140	OK	0.00	-0.2945	0.0140
ex6_2_10	-3.0520	OK	0.01	-3.0520	7.82e-04	OK	0.00	-3.0520	7.82e-04
ex6_2_11	0.0000	OK	0.02	-2.672e-6	3.00e-06	OK	0.00	-2.672e-6	3.00e-06
ex6_2_12	0.2892	OK	0.01	0.2892	0.0018	OK	0.00	0.2892	0.0018
ex6_2_13	-0.2162	OK	0.00	-0.2162	0.0044	OK	0.00	-0.2162	0.0044
ex6_2_14	-0.6954	OK	0.01	-0.6954	0.0060	OK	0.01	-0.6954	0.0060
ex6_2_5	-70.7521	OK	0.01	-70.7521	3.10e-05	OK	0.00	-70.7521	3.10e-05
ex6_2_6	0.0000	OK	0.00	7.108e-7	1.00e-06	OK	0.00	7.108e-7	1.00e-06
ex6_2_7	-0.1608	OK	0.01	-0.1608	0.0296	OK	0.00	-0.1608	0.0296
ex6_2_8	-0.0270	OK	0.00	-0.0270	0.0235	OK	0.00	-0.0270	0.0235
ex6_2_9	-0.0341	OK	0.01	-0.0341	0.0992	OK	0.01	-0.0341	0.0992
ex7_2_1	1227.1896	OK	0.03	1227.2254	0.0029	OK	0.01	1227.2254	0.0029
ex7_2_10	0.1000	OK	0.05	0.1000	2.50e-04	ITER	0.79	0.1034	3.40
ex7_2_2	-0.3888	OK	0.01	-0.3888	0.0029	OK	0.00	-0.3888	0.0029
ex7_2_3	7049.2181	OK	0.02	7049.2476	4.19e-04	OK	0.00	7049.2476	4.19e-04
ex7_2_4	3.9180	OK	0.02	3.9180	2.59e-04	OK	0.09	3.9180	2.59e-04
ex7_2_5	1.012e+4	OK	0.02	1.012e+4	1.02e-04	OK	0.00	1.012e+4	1.02e-04
ex7_2_6	-83.2499	OK	0.02	-83.2497	2.05e-04	OK	0.00	-83.2497	2.05e-04
ex7_2_7	-5.7399	OK	0.01	-5.7398	0.0014	OK	0.00	-5.7398	0.0014
ex7_2_8	-6.0820	OK	0.02	-6.0820	1.67e-04	OK	0.03	-5.8169	4.36
ex7_2_9	1.1436	OK	0.03	1.1436	0.0020	OK	0.01	1.1436	0.0020
ex7_3_1	0.3417	OK	0.03	0.3417	0.0116	OK	0.01	0.3417	0.0116
ex7_3_2	1.0899	OK	0.01	1.0899	0.0033	OK	0.00	1.0899	0.0033
ex7_3_3	0.8175	OK	0.01	1.1448	40.03	OK	0.00	1.1448	40.03
ex7_3_4	6.2746	ERR	0.01	0.0000	100.00	OK	0.01	10.0000	59.37

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(continuacao da Tabela 2)

		Ipopt				**MadNLP**			
Problema	F_{best}	status	tempo	obj.	gap (%)	status	tempo	obj.	gap (%)
ex7_3_5	1.2036	ERR	0.01	0.0000	100.00	OK	0.01	1.2069	0.2739
ex7_3_6	0.0000	INV	0.00	0.0000	0.00e+00	INV	0.11	0.0000	0.00e+00
ex8_1_1	-2.0218	OK	0.01	-2.0218	3.36e-04	OK	0.00	-2.0218	3.36e-04
ex8_1_2	-1.0709	OK	0.01	-0.7970	25.58	OK	0.00	-0.7970	25.58
ex8_1_3	3.0000	OK	0.01	30.0000	900.00	OK	0.00	30.0000	900.00
ex8_1_4	0.0000	OK	0.00	0.0000	0.00e+00	OK	0.00	0.0000	0.00e+00
ex8_1_5	-1.0316	OK	0.00	0.0000	100.00	OK	0.00	0.0000	100.00
ex8_1_6	-10.0860	OK	0.01	-5.0654	49.78	OK	0.00	-5.0654	49.78
ex8_1_7	0.0293	OK	0.03	0.0293	0.0370	OK	0.01	0.0293	0.0370
ex8_1_8	-0.3888	OK	0.01	-0.3888	0.0029	OK	0.01	-0.3888	0.0029
ex8_2_1	-979.1783	OK	0.04	-979.1783	0.00e+00	OK	0.01	-979.1783	0.00e+00
ex8_2_4	-5.831e+7	ERR	2.25	2.650e+10	45547.41	ITER	3.57	0.0232	100.00
ex8_3_1	-0.7937	OK	0.74	-0.8196	3.26	OK	0.11	-0.8196	3.26
ex8_3_10	-0.8845	OK	0.24	-1.3193	49.16	OK	0.10	-1.3193	49.16
ex8_3_12	-0.5825	OK	0.37	-0.9906	70.06	OK*	0.06	-2.521e-5	100.00
ex8_3_13	-42.3904	OK	0.32	-41.9181	1.11	OK	0.07	-36.0847	14.88
ex8_3_14	-2.4124	OK	1.07	-2.4977	3.54	OK	0.12	-2.4986	3.57
ex8_3_2	-0.4123	OK	0.03	-0.3313	19.64	OK	0.02	-0.3313	19.64
ex8_3_3	-0.4166	OK	0.02	-0.3330	20.06	OK	0.02	-0.3330	20.06
ex8_3_4	-3.5800	OK	0.20	-3.5800	4.93e-04	INF	0.48	-3.5501	0.8347
ex8_3_5	-0.0688	OK	0.14	-0.0691	0.4646	OK	0.09	-0.0691	0.4645
ex8_3_6	-0.5000	OK	0.06	-0.5000	0.00e+00	OK	0.02	-0.5000	0.00e+00
ex8_3_7	-1.2326	OK	0.45	-1.2326	0.0016	OK	0.16	-1.2326	0.0016
ex8_3_8	-3.2468	OK	0.30	-3.2561	0.2870	OK	0.20	-3.2561	0.2870
ex8_3_9	-0.7630	OK	0.03	-0.7416	2.80	OK	0.01	-0.7416	2.80
ex8_4_1	0.6186	OK	0.01	0.6186	0.0044	OK	0.00	0.6186	0.0044
ex8_4_2	0.4852	OK	0.01	0.4852	0.0098	OK	0.00	0.4852	0.0098
ex8_4_3	0.0046	OK	0.01	0.0046	1.08	OK	0.00	0.0046	1.08
ex8_4_4	0.2125	OK	0.01	0.2125	0.0189	OK	0.00	0.2125	0.0189
ex8_4_5	3.000e-4	OK	0.02	0.0012	308.43	OK	0.01	0.0012	308.43
ex8_4_6	0.0011	OK	0.09	0.6590	59805.84	OK	0.01	0.0011	0.4526
ex8_4_7	28.8980	OK	0.01	29.0473	0.5167	OK	0.00	29.0473	0.5167
ex8_4_8	3.3218	OK	0.03	3.3218	0.0014	OK	0.02	3.3218	0.0014
ex8_5_1	0.0000	OK	0.01	-4.072e-7	0.00e+00	OK	0.00	-4.072e-7	0.00e+00
ex8_5_2	-2.000e-4	OK	0.03	-1.150e-8	99.99	OK	0.02	-1.150e-8	99.99
ex8_5_3	-0.0042	OK	0.02	-0.0041	1.54	OK	0.01	-0.0041	1.54
ex8_5_4	-4.000e-4	OK	0.01	-4.251e-4	6.29	OK	0.00	-4.251e-4	6.29
ex8_5_5	-0.0108	OK	0.01	6.567e-7	100.01	OK	0.00	6.567e-7	100.01
ex8_5_6	-0.0012	OK	0.01	9.881e-7	100.08	OK	0.00	9.881e-7	100.08
ex8_6_1	-28.4225	OK	0.24	-28.4225	1.12e-04	OK	0.12	-27.5559	3.05
ex8_6_2	-31.8886	OK	0.09	-31.8886	9.30e-05	OK	0.06	-31.8886	9.30e-05
ex9_1_1	-13.0000	OK*	0.02	-12.9601	0.3067	OK	0.01	-13.0000	0.00e+00
ex9_1_10	-3.2500	OK	0.08	-3.2500	0.00e+00	INF	0.03	-3.2443	0.1767
ex9_1_2	-16.0000	OK	0.04	-16.0000	0.00e+00	OK	0.01	-16.0000	0.00e+00
ex9_1_3	-58.0000	OK	0.01	-58.0000	2.00e-06	OK	0.01	-58.0000	2.00e-06
ex9_1_4	-37.0000	OK	0.03	-37.0000	0.00e+00	OK	0.01	-37.0000	0.00e+00
ex9_1_5	-1.0000	OK	0.02	4.0000	500.00	OK	0.01	4.0000	500.00
ex9_1_6	-52.0000	OK	0.01	-52.0000	0.00e+00	OK	0.00	-52.0000	0.00e+00
ex9_1_7	-50.0000	OK	0.01	-50.0000	2.00e-06	OK	0.00	-50.0000	2.00e-06
ex9_1_8	-3.2500	OK	0.11	-3.2500	0.00e+00	INF	0.07	-3.2443	0.1767
ex9_1_9	2.0000	OK	0.01	2.0000	0.00e+00	OK	0.00	2.0000	0.00e+00

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(continuacao da Tabela 2)

Problema	F_{best}	Ipopt status	tempo	obj.	gap (%)	MadNLP status	tempo	obj.	gap (%)
ex9_2_1	17.0000	OK	0.02	25.0000	47.06	OK	0.00	25.0000	47.06
ex9_2_2	99.9995	OK*	0.03	100.0000	5.00e-04	OK*	0.01	100.0000	5.00e-04
ex9_2_3	0.0000	OK	0.03	5.0000	5.00	OK	0.01	5.0000	5.00
ex9_2_4	0.5000	OK	0.01	0.5000	0.00e+00	OK	0.01	0.5000	0.00e+00
ex9_2_5	5.0000	OK	0.02	9.8000	96.00	OK	0.01	9.8000	96.00
ex9_2_6	-1.0000	OK	0.06	-1.0000	0.00e+00	OK	0.03	-1.0000	0.00e+00
ex9_2_7	17.0000	OK	0.02	25.0000	47.06	OK	0.01	25.0000	47.06
ex9_2_8	1.5000	OK	0.01	1.5000	1.00e-06	OK	0.00	1.5000	1.00e-06
gtm	543.5651	OK	0.02	543.5651	1.00e-06	OK	0.01	543.5651	1.00e-06
harker	-986.5135	OK	0.01	-986.5135	2.00e-06	OK	0.02	-986.3677	0.0148
haverly	-400.0000	OK	0.02	-400.0000	1.00e-06	OK	0.00	-400.0000	1.00e-06
hhfair	-87.1590	OK	0.02	-87.1590	4.30e-05	OK	0.01	-87.1590	4.30e-05
himmel11	-3.067e+4	OK	0.01	-3.067e+4	2.00e-06	OK	0.00	-3.067e+4	2.00e-06
himmel16	-0.8660	OK	0.02	-0.6750	22.06	OK	0.01	-0.6750	22.06
house	-4500.0000	OK	0.02	-4500.0000	1.00e-06	OK	0.04	-4500.0000	1.00e-06
hydro	4.367e+6	OK	0.01	4.367e+6	2.00e-06	OK	0.01	4.367e+6	2.00e-06
immun	0.0000	OK	0.01	1.138e-19	0.00e+00	OK	0.01	3.524e-18	0.00e+00
launch	2257.7976	OK	0.04	2257.7973	1.20e-05	OK	0.02	2257.7973	1.20e-05
least	1.409e+4	OK	0.01	1.409e+4	0.00e+00	OK	0.00	1.409e+4	0.00e+00
linear	89.0000	NUM	0.10	503.4724	465.70	ITER	1.41	89.2951	0.3315
lnts100	0.5546	OK	0.04	0.5546	8.29e-04	OK	0.48	0.5546	8.29e-04
lnts50	0.5547	OK	0.02	0.5547	0.0056	OK	0.02	0.5547	0.0056
meanvar	5.2434	OK	0.00	5.2434	1.90e-05	OK	0.00	5.2434	1.90e-05
mhw4d	0.0293	OK	0.01	27.8719	95025.96	OK	0.00	27.8719	95025.96
minlphi	582.2361	OK	0.01	582.2361	7.00e-06	OK	0.01	582.2361	7.00e-06
otpop	0.0000	OK	0.03	2.128e-18	0.00e+00	OK	0.02	2.796e-19	0.00e+00
pindyck	-1612.1783	OK	0.01	-1170.4863	27.40	OK	0.01	-1170.4863	27.40
pollut	-5.353e+6	OK	0.02	-5.353e+6	0.00e+00	OK	0.01	-5.353e+6	1.00e-06
polygon100	-0.6791	OK	2.00	-0.7269	7.03	OK	2.42	-0.7269	7.03
polygon25	-0.7722	OK	0.14	-0.7748	0.3352	OK	0.11	-0.7748	0.3352
polygon50	-0.7757	OK	0.42	-0.7197	7.21	OK	0.40	-0.7197	7.21
process	-1161.3366	OK	0.00	-1161.3366	1.00e-06	OK	0.00	-1161.3366	1.00e-06
prolog	0.0000	OK	0.07	4.604e-7	0.00e+00	OK	0.01	4.619e-7	0.00e+00
qp1	8.000e-4	OK	0.12	8.093e-4	1.17	OK	0.10	8.094e-4	1.18
qp2	8.000e-4	OK	0.05	8.093e-4	1.17	OK	0.05	8.094e-4	1.18
qp3	8.000e-4	OK	0.02	8.094e-4	1.18	OK	0.01	8.094e-4	1.18
qp4	8.000e-4	OK	0.01	8.094e-4	1.18	OK	0.01	8.094e-4	1.18
qp5	0.4315	OK	0.02	0.4315	0.0106	OK	0.01	0.4315	0.0106
ramsey	-2.4875	OK	0.02	-2.4875	0.0013	OK	0.01	-2.4875	0.0013
rbrock	0.0000	OK	0.02	2.908e-19	0.00e+00	OK	0.00	2.905e-19	0.00e+00
robot50	9.1469	NO NL	–	–	–	NO NL	–	–	–
rocket100	-1.0126	OK	0.35	-1.0128	0.0229	OK	0.23	-1.0128	0.0229
rocket50	-1.0128	OK	0.21	-1.0128	0.0017	OK	0.19	-1.0128	0.0017
sambal	3.9682	OK	0.00	3.9682	5.13e-04	OK	0.02	136.3722	3336.63
sample	726.6367	OK	0.01	726.6789	0.0058	OK	0.00	726.6789	0.0058
ship	0.0000	OK	0.01	5.5409	5.54	OK	0.01	5.5409	5.54
srcpm	2109.7818	OK	0.01	2109.7818	1.00e-06	OK	0.00	2109.7818	1.00e-06
turkey	-2.933e+4	OK	0.26	-2.933e+4	0.00e+00	OK	0.20	-2.933e+4	0.00e+00
wall	-1.0000	OK	0.01	1.0000	200.00	OK	0.00	1.0000	200.00
water	906.3519	OK	0.08	1001.1633	10.46	OK	0.02	1079.4248	19.10
weapons	-1735.5700	OK	0.02	-1735.5696	2.40e-05	OK	0.01	-1735.5696	2.40e-05